

Pumping theorem

pumping length



Let L be a regular lang. There exists an integer $p \geq 1$ s.t any string $w \in L$ such that $|w| \geq p$ can be divided into three pieces $w = xyz$, satisfying the follow conditions.

1. for each $i \geq 0$, $xy^i z \in L$

2. $|y| > 0$

3. $|xy| \leq p \Rightarrow |y| \leq p$

Example

$L = ab^*a$ $p=3$

$\begin{matrix} a & b & b & a \\ \underbrace{}_x & \underbrace{}_y & \underbrace{}_z & \end{matrix}$

$xy^0 z = aba \in L$

$xy^2 z = abbb a \in L$

$xy^i z = ab^{i+1} a \in L$

Proof.

Let M be a DFA that accepts L .

Let p be # distinct states of M .

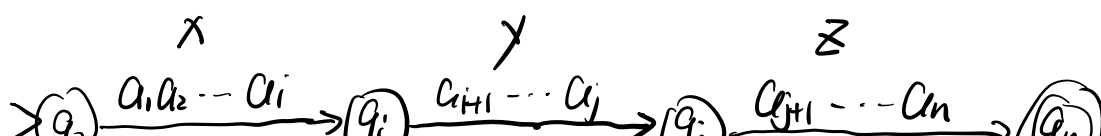
If every string in L has length less than p , // L is finite trivial

otherwise let w be any string in L with $|w| \geq p$.

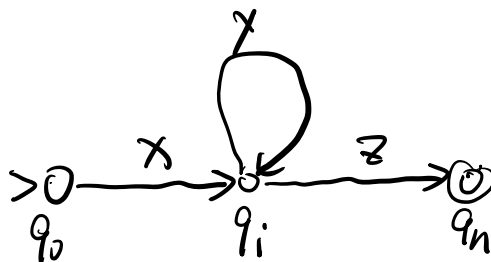
$w = a_1 a_2 \dots a_n$ $n \geq p$



there must $0 \leq i < j \leq p$ such that $q_i = q_j$



1. $\forall t \geq 0, xy^t z \in L$
2. $|y| > 0 \quad |y| = j - i > 0$
3. $|xy| = j \leq p$



□

regular $\Rightarrow$ pumping

pumping $\nRightarrow$ regular

pumping $\Rightarrow$ non-regular

Example

Show that $L = \{a^i b^i : i \geq 0\}$ is not regular.

Proof.

Assume for the sake of contradiction, L is regular.

Let p be the pumping length.

Let $w = a^p b^p \in L$. w can be written as $w = xyz$ such that

(1) for $i \geq 0, xy^i z \in L$

(2) $|y| > 0$

(3) $|xy| \leq p$

$\underbrace{a \dots a}_p \underbrace{b \dots b}_p$
 $(2)(3) \Rightarrow y = a^k \text{ for some } k \geq 1$

$xy^0 z = a^{p-k} b^p \notin L$.

contradicting with (1).

Example

Show $L = \{ w \in \{a, b\}^* : w \text{ has equal number of } a\text{'s and } b\text{'s} \}$ is not regular.

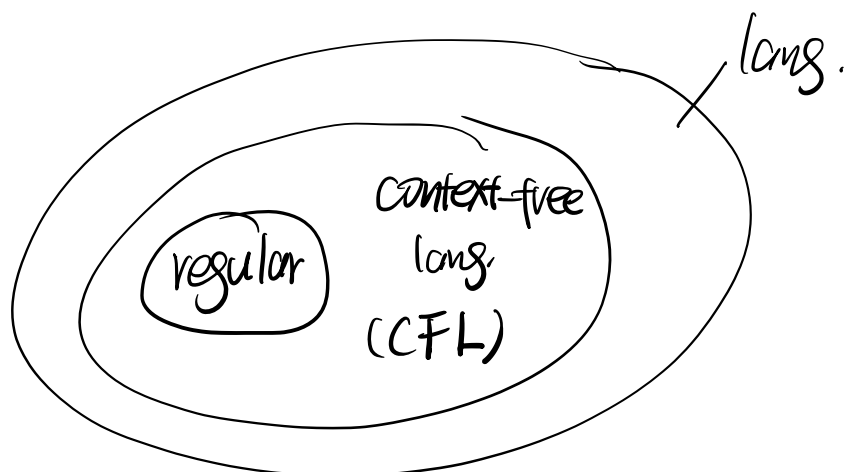
Proof.

Assume that L is regular.

$L \cap a^*b^*$ must be regular

||

$\{ a^i b^i : i \geq 0 \}$ contradiction!



1. generator: context-free grammar (CFG) ↗
2. recognizer: pushdown automata (PDA) ↙
3. pumping theorem for CFL

Example

$S \rightarrow aSb$

rule

$S \rightarrow A$

a, b : terminal

$A \rightarrow c$

S, A : non-terminal

S : start symbol

$$S \Rightarrow aSb \Rightarrow aaSbb \Rightarrow aaAbb \Rightarrow aacbb$$

Definition

A context-free grammar (CFG) is 4-tuple $G = (V, \Sigma, R, S)$

- V is a set of symbols
- $\Sigma \subseteq V$ is the set of terminals; $V - \Sigma$ " " non-terminals.
- $S \in V - \Sigma$ is the start symbol
- R , the set of rules, is a *finite* subset of $(V - \Sigma) \times V^*$.

$$(A, u)$$

$$A \xrightarrow{G} u$$

for any $x, y, u \in V^*$, for any $A \in V - \Sigma$,

we say $xAy \xrightarrow{G} xuy$ if $(A, u) \in R$
 $\downarrow$
 derives in one step.

for $x, y \in V^*$,

we say $x \xRightarrow{*}_G y$ if $x = y$ or $\exists w_0, w_1, \dots, w_n \in V^*$ s.t
 derives

$$x = w_0$$

$$w_{i-1} \xrightarrow{G} w_i \text{ for } i \in [1, n]$$

$$w_n = y$$

$$w_0 \Rightarrow w_1 \Rightarrow w_2 \Rightarrow \dots \Rightarrow w_n$$

derivation from w_0 to w_n

of length n

We say G generates a string $w \in \Sigma^*$ if $S \xRightarrow{*}_G w$.

$L(G) = \{w \in \Sigma^* : G \text{ generates } w\}$
 $G \text{ generates } L(G)$

Definition (CFL)

A language is context-free if some CFG generates it.

Exercise

Show that $\{w \in \{a,b\}^* : w = w^R\}$ is a CFL.

1. $|w|=0$ $w=e$
2. $|w|=1$ $w=a, w=b$
3. $|w| \geq 2$, $w = aua, w = bu b, u = u^R$

$S \rightarrow e$
 $S \rightarrow a$
 $S \rightarrow b$
 $S \rightarrow aSa$
 $S \rightarrow bSb$

$S \rightarrow e|a|b|aSa|bSb$

if $w \in L(G)$, $w = w^R$

by induction on # steps of derivations of w

if $w = w^R$, $w \in L(G)$

by induction on length of w

Example

$\{w \in \{(,)\}^* : w \text{ is a string of balanced parentheses}\}$ is a CFL.

$$S \rightarrow SS$$

$$S \rightarrow (S)$$

$$S \rightarrow e$$

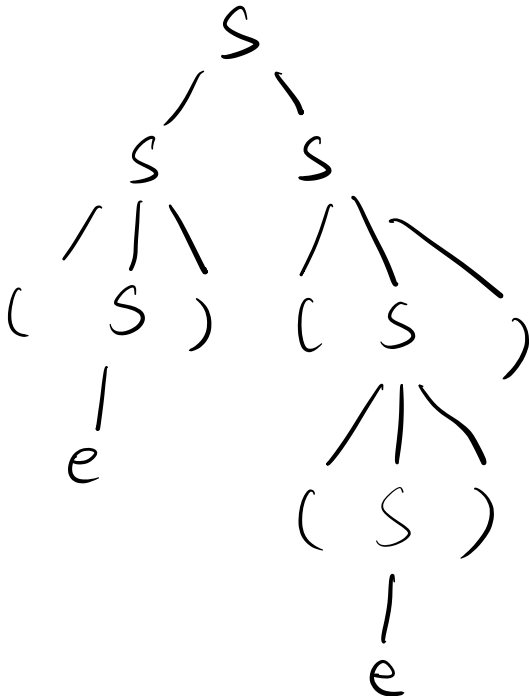
$()(())$

leftmost derivation

$$S \Rightarrow SS \Rightarrow (S)S \Rightarrow ()S \Rightarrow ()(S) \Rightarrow ()((S)) \Rightarrow ()(())$$

$$S \Rightarrow SS \Rightarrow S(S) \Rightarrow S((S)) \Rightarrow S(()) \Rightarrow (S)(()) \Rightarrow ()(())$$

rightmost " "



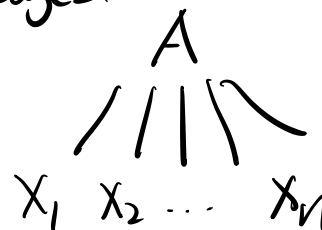
parse tree

internal nodes: non-terminals

leaves: terminals or e

e must be only child of its parent

edges:



$$A \rightarrow X_1 \dots X_n$$

Example

$$E \rightarrow E + E$$

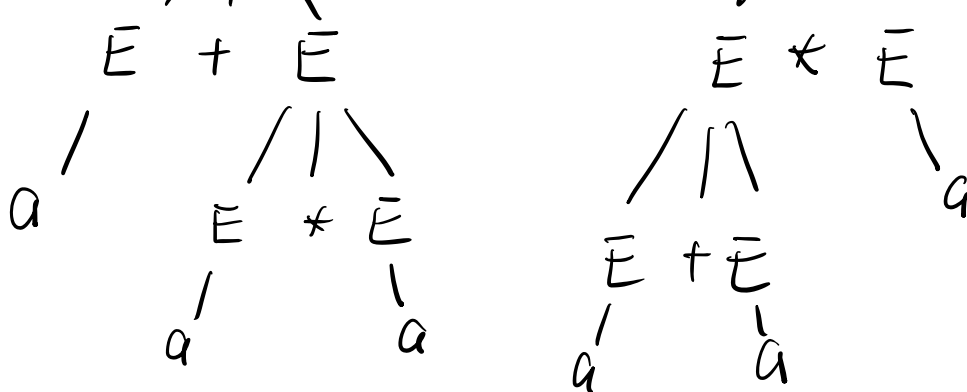
$$E \rightarrow E * E$$

$a + a * a$



$$\bar{E} \rightarrow (E)$$

$$\bar{E} \rightarrow a$$



Definition

A CFG G is ambiguous if for $w \in L(G)$,

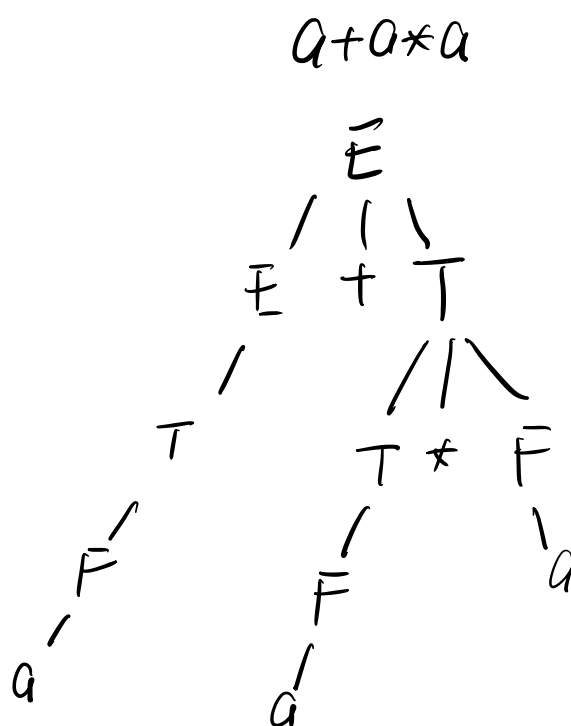
there two or more parse trees with root S and yield w .

Example

$$E \rightarrow E + T \mid T$$

$$T \rightarrow T * F \mid F$$

$$F \rightarrow (E) \mid a$$



Fact

Some CFL is inherently ambiguous. That is, every CFG that generates this language is ambiguous.

$$\{a^i b^j c^k : i=j \text{ or } j=k\}$$

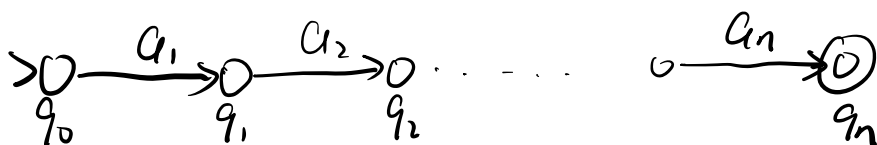
$$a^4 b^4 c^4$$

Theorem

Every regular language is context-free.

DFA $M \longrightarrow$ CFG G s.t. $L(G) = L(M)$

$$w = a_1 \dots a_n$$



$$\begin{aligned} q_0 &\rightarrow a_1 q_1 \rightarrow a_1 a_2 q_2 \rightarrow a_1 a_2 a_3 q_3 \rightarrow \dots \rightarrow a_1 a_2 \dots a_n q_n \\ &\rightarrow a_1 a_2 \dots a_n \end{aligned}$$

$$M = (K, \Sigma, \delta, s, F)$$

$$G = (V, \Sigma, R, S)$$

$$V = K \cup \Sigma$$

$$S = s$$

$$R = \{ q \rightarrow ap : \delta(q, a) = p \} \cup \{ q \rightarrow e : q \in F \}$$

$$\forall w \in \Sigma^*$$

$$(s, w) \vdash_M^* (q, e) \text{ for some } q \in F$$

... γ ...

$$\Leftrightarrow S \Rightarrow_G w$$